Validation Windows: Epistemic Uncertainty Produced by the Solver in a Literature-Derived Bayesian Network

Deborah Vakas Duong [1] Igor Yi [1]

Abstract

When a Bayesian network is built from published scientific studies, each contributing a point estimate and a 95% confidence interval, the aggregation problem is fundamentally epistemic, not just statistical. Different studies on the same edge disagree; different edges imply joint probabilities that must satisfy the law of total probability; and the resulting conditional probability tables (CPTs) must also be consistent with observed population frequencies. We describe a system that solves the aggregation as a quadratic program whose feasible region is a convex polytope of CPTs consistent with all sources of evidence simultaneously. The width of the polytope along each study's axis, which we call the validation window, is a first-class representation of epistemic uncertainty for that edge: produced by the construction itself, not computed post-hoc as a calibration layer. It is tight where literature and population data agree, wide where they disagree, and exceeds the study's own literature CI when the solver must stretch the study to achieve global consistency. The system sits between two transformer endpoints: an upstream LLM that extracts literature into structured spreadsheet rows (making the pipeline domain-portable), and a downstream multi-agent coevolutionary simulation it seeds with behaviors. On a 427-node longevity-risk network built from 166 published studies (72% meta-analyses, $\sim 4\times 10^8$ pooled subjects), the median window shrinks by 87% (from 0.125 to 0.016) as consistent literature-established edges are added. A residual set of low-base-rate edges where the solver cannot commit to a direction, the $\approx 2\times 10^{-4}$ pancreatic-cancer chain, demonstrates the intended failure mode: abstention, visible as inversion of the CPT direction precisely where literature alone cannot discipline the joint. Beyond the window, the same construction is usable as a predictor: with each target's definitional-surrogate biomarkers excluded, it discriminates held-out clinical events at a mean per-respondent ROC AUC of 0.72 across 16 targets (up to 0.94 on coronary artery disease), at or above the level of cohort-fitted clinical risk calculators on the better-instrumented targets. The window thus provides a direct, per-edge measure of what the model does not know, computed by the construction itself.

1. Introduction

A Bayesian network built from the published scientific literature faces three simultaneous epistemic challenges. Individual uncertainty: each primary or meta-analytic study reports a point estimate and an interval, not a certainty. Cross-study disagreement: two studies on the same edge often report different effect sizes with overlapping but non-identical intervals. Joint coherence: a collection of marginal effect sizes, assembled into a network, must respect the law of total probability and must remain consistent with population-level frequencies from a reference survey dataset. Collapsing all of this to a single CPT per node, the default in published expert systems (Heckerman, 1998), discards most of the epistemic information before it reaches the user. Returning the set of CPTs consistent with all sources, by contrast, preserves the shape of what is not known.

Transformers sit on both ends of the pipeline we describe. Upstream, a large language model (Brown

[1] Agent Based Learning Systems, San Luis Obispo, California, USA. Correspondence to: Deborah Vakas Duong <dduong@agentBasedLearningSystems.com>.

2nd Workshop on Epistemic Intelligence in Machine Learning (EIML@ICML 2026), Seoul, South Korea.

et al., 2020) reads published meta-analyses and primary studies, extracting effect sizes, confidence intervals, populations, and citations into the structured spreadsheet rows our solver consumes. This LLM-mediated extraction is what makes the method generalizable: any scientific domain with peer-reviewed effect-size literature, longevity and health in our demonstration, but also macroeconomic policy, education interventions, or criminal-justice evaluation, can be converted to a network without domain-specific hand-curation. Downstream, once the network carries treatment (interventional) edges it is itself a policy net that seeds the agents of a multi-agent coevolutionary simulation with behaviors.

This paper presents the network-construction step. The system takes as input a spreadsheet of literature-sourced edges (effect sizes plus 95% confidence intervals) together with a reference dataset of population covariate frequencies (we use NHANES (Centers for Disease Control and Prevention, National Center for Health Statistics, 2020) throughout), and produces a Bayesian network whose CPTs are the solution of a quadratic program. The QP's objective is heuristic, a least-squares fit to the literature point estimates plus a monotonicity penalty that discourages direction inversions, and its constraints are probability-law: the CPT is a valid probability table, each literature RR lies within a windowed band, and the law of total probability holds across groups of cells. We call the last item the subset constraints. The feasible region of the constrained QP is a convex polytope of CPTs consistent with all sources of evidence simultaneously.

Our contribution is conceptual and algorithmic: a working quadratic-program solver that constructs the polytope from literature in a single pass, together with the argument that the width of the feasible polytope along each study's axis, the validation window, is the quantity the system already computes that directly answers the core epistemic-uncertainty question: for this edge, how much does the model not know? Tight windows mean "literature and population data mutually confirm this effect under the structure I've built." Wide windows mean "I had to bend the literature to make the whole thing fit." Windows that exceed the raw literature CI mean "my structure may be wrong; attend here." The system's operating loop iteratively searches for structural changes or additional literature that narrows the widest windows; the collective width of the polytope serves as the monotone-decreasing measure of residual epistemic uncertainty. When the network is handed off to downstream transformer agents, these windows travel with each edge: the transformer receives a prior that makes its own uncertainty commensurate with the literature's. As supplementary evidence that these literature-constrained CPTs carry real predictive structure, the same network discriminates held-out clinical events at a mean per-respondent ROC AUC of 0.72 (up to 0.94), with diagnostic biomarkers excluded (App. F).

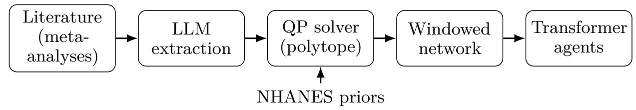

Figure 1. The end-to-end pipeline. An LLM extracts published effect sizes into structured rows; the QP solver returns the feasible polytope of CPTs; each edge carries its validation window into the downstream transformer agents. NHANES supplies per-node priors to the solver.

The system is not an imprecise-probability engine (§3 relates the feasible polytope to credal sets (Walley, 1991; Cozman, 2000)). It is not a calibration (Guo et al., 2017) or conformal-prediction (Vovk et al., 2005) overlay on a pre-trained classifier. It is a construction process whose intermediate artifact, the windowed polytope, is the epistemic representation. The construction itself reports the residual uncertainty on each edge.

2. The System End-to-End

Figure 1 summarises the pipeline end to end; the three construction stages (Inputs, Compilation, CPT solve) are described in turn.

Inputs. A single spreadsheet. A study row carries an output node (e.g. lung_cancer), an input node and its value (smoking_yes), a statistic type (RR, OR, HR, or effect size), a point estimate, a $\pm$ for the 95% CI, and a citation; our demonstration network has 166 such rows, 72% of which are meta-analyses, with a pooled total of $\sim 4.17 \times 10^8$ subjects. A definition row declares a node's type: a directly-observed NHANES variable, a dependency node whose CPT the QP solves from its parents' study RRs, or a logical aggregator (any_of, all_of) combining parents. Still other rows declare a link between nodes: a subset relation (is_a, subsumes) or identity (equivalent_to) for definitional containment, or a distal link routing a computed aggregator-level effect onto a disease. The build also takes the reference dataset NHANES 2017–March 2020, 116K respondents.

Compilation. The spreadsheet compiles to a DAG with one node per output and one edge per (input, output) pair. Aggregator definitions resolve to deterministic truth tables. Distal links, 33 in our demonstration, organizing leaves into sub-aggregators that feed a disease, are handled by a Monte-Carlo procedure that propagates study-level RRs through the aggregator's NHANES-weighted co-occurrence structure to produce a single aggregator-to-disease RR with its own 95% CI.

CPT solve. For each node Y with parents $X_1 \dots X_K$, let $\mathbf{x}$ be the flattened CPT. Each literature study i bearing on Y gives four linear constraints $r_k^{(i)}\mathbf{x} = s_k^{(i)}$ (the standard marginal-RR expansion): each $r_k^{(i)}$ expresses study i's marginal RR as a linear functional of the flattened CPT via the law of total probability, fixed by the study's (parent, value, target) and the reference-survey marginal, not a free choice. Stacking these rows gives the matrix R (the collection of the $r_k^{(i)}$) and target vector $\mathbf{s}$. The QP objective (both terms heuristic) is

$$\mathcal{L}(\mathbf{x}) = \|R\mathbf{x} - \mathbf{s}\|_2^2 \;+\; \lambda \sum_p \max(0, \mathrm{inv}_p)^2, \qquad (1)$$

where the first term fits the literature point estimates and the second penalises monotonicity inversions, with weight $\lambda > 0$ (p indexes signed-direction (parent, value) pairs and inv_p is the violation amount, zero when the direction holds). We do not claim this objective follows from a Bayesian or decision-theoretic principle; it is a simple heuristic that merely selects one interior point of the polytope. The probability laws live in the constraints, not the objective, so the reported window W depends only on the constraint polytope (§3, App. A) and is invariant to λ. The constraints that encode probability laws are: (i) $\mathbf{x}$ is a valid probability table (row sums, non-negativity); (ii) a windowed literature band per study, $s - c \le R\mathbf{x} \le s + c$, with c scaled per-study by the study's own literature-CI width (narrower CIs yield tighter effective bounds) and the global scale determined by a binary search on feasibility; (iii) subset constraints, a set of linear equalities/inequalities requiring that, for every group of CPT cells linked by the law of total probability, the corresponding relation holds within the windowed tolerance. Subset constraints range over groups of cells induced by the CPT's full structure, including marginalisation over any subset of parents. Grouping the terms this way isolates the heuristic part of the solve (literature-fit and monotonicity) from the probability-law part (the constraints). Least squares and monotonicity are retained as soft heuristics because marginal effect sizes and their directions are known properties of the literature, not probability laws. NHANES supplies per-node priors through the definition rows; it does not supply conditional training signal to the solver, which would create a circularity in the evaluation metrics below.

The validation window W is a confidence interval. The polytope's boundary is determined by the probability-law constraints, not by the objective; W is therefore independent of λ. The window c is not a free parameter, it is determined by a binary search on feasibility. We start with c=1.0, which trivially admits any probability table, and shrink c toward 0 as far as the constrained QP remains feasible. The smallest feasible c is recorded per edge as the edge's validation window, $W \in [0, 1]$, in the same probability units as the CPT cells it bounds. W=0 means the constraints uniquely determine the edge's CPT row; W=1 means the constraints are non-binding.

Under the normal approximation, each input 95% CI is a likelihood-ratio confidence region at level α=0.05, so W_e is the half-width of a confidence interval for the edge's CPT cell and the solver reports a per-edge CI coverage cov_e alongside W_e. Coverage scales with the input-band factor c via $c \cdot z_{1-\alpha/2} = z_{1-\alpha/(2n)}$ (Bonferroni); for n=166 studies, joint coverage 0.95 corresponds to $c{\approx}1.85$. Runs in this paper use c=1; intervals are exact marginally and conservative jointly, and can be re-generated at any target joint coverage via the Bonferroni relation. When studies on different edges are mutually consistent, W collapses toward the literature CI width; when they are not, c must widen past the literature CI to avoid infeasibility, and the amount by which W exceeds the literature CI is the system's explicit complaint.

3. Windows as Epistemic Uncertainty

The feasible polytope of CPTs satisfying all constraints at the minimum-feasible c is a convex region (Fig. 2). Along each study axis its extent is $[L_i, U_i]$, and $W_i = U_i - L_i \in [0, 1]$ is the validation window for that study. By construction this is a projected credal set in the sense of Walley (1991) and Cozman (2000), but arrived at constructively from published literature rather than from elicited lower and upper probabilities. Four properties make W a useful epistemic report:

(1) Monotone in evidence. Adding a consistent literature edge can only tighten the polytope; an inconsistent edge is detected because c must widen (or the QP becomes infeasible). In our experiments the median window $\tilde{W}$ dropped by 87% (from 0.125 to 0.016) as thirteen structural edges from a medical-knowledge audit were added, meaning that by the final build, half of all solved edges have $W \le 0.016$ (their literature +

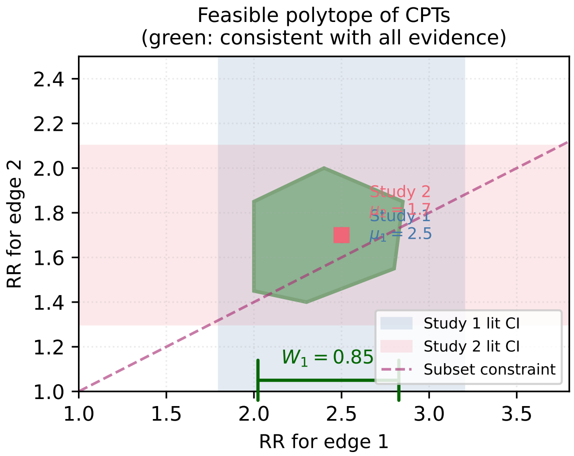

Figure 2. Feasible polytope of CPTs consistent with two literature studies (blue, pink bands) and a law-of-total-probability subset constraint (dashed magenta). The green polygon is the solution set; its projection onto the study-1 axis yields the validation window W_1. When the projection equals the literature CI, the system has extracted no additional constraint; when it is narrower, cross-edge coherence has tightened the prior; when it exceeds the literature CI, the solver has stretched the study, the system's explicit complaint that something else in the network is in tension with this study.

subset-constraint content is almost fully binding), and the remaining half spread up to 1.0.

(2) Localizes cross-edge tension. The system sorts studies by W. Studies with high W are edges where the per-study literature CI disagrees with the joint of all other constraints in the network: either literature on this edge contradicts literature on an adjacent edge, or literature contradicts population data, or the DAG is missing a structural edge that would resolve the tension. In the demonstration network the top-20 window list after each iteration provided the to-do list for the next iteration. The three edges that remain stuck at $W{=}1$ after all available evidence has been added are the diabetes/healthy-diet pairs; their literature pulls in a direction that no current structural change has reconciled with the adjacent constraints.

(3) Exceeds the literature CI when forced. A window wider than the study's own literature CI is a report by the system that something upstream is wrong: the chosen structure, another study, or a NHANES leaf mapping.

(4) Determines untested parent combinations. For K parents the CPT has 2^K rows; literature typically measures only the K marginal effects. Given the marginals and the reference survey's parent co-occurrence structure, the subset constraints mathematically determine the remaining $2^K - K$ rows to within the window, combinations no study has jointly examined, provided the co-occurrence exceeds a threshold we make precise in App. B. We measure co-occurrence with a signed deviation, $\rho(X_i, X_j) = \max_{v_i}\big[P(X_j = \hat{x}_j \mid X_i = v_i) - P(X_j = \hat{x}_j)\big]$, with $\hat{x}_j$ the parent's reference state and the sign of the maximising deviation retained. Because the derivation is the law of total probability, these predictions need no further experiment. The required $|\rho|$ is modest; where it is not yet sufficient, the objective returns the best estimate given the known marginals and the DAG, with the window reporting the residual uncertainty.

A complementary abstention signature (§5) is direction symmetry within a small feasible polytope: W can be modest yet the solver cannot pick a committed direction. That failure does not surface in W; it surfaces in the query-RR direction check.

4. Empirical Demonstration

Setup. Our demonstration network covers longevity risk: 427 nodes (227 epidemiological variables from NHANES plus 200 structural nodes including 15 disease targets, 33 distal aggregators, and 153 definitional sub-variables); 166 study rows (72% meta-analyses, $\sim$4.17$\times$10^8 pooled subjects); a 116K-respondent NHANES reference set. We evaluate three things per build:

- Direction accuracy: for each study, clamp its input at yes, query its target, compare the sign of the resulting query RR to the literature RR's sign; report the fraction correct out of 369 testable studies.
- Median window $\tilde{W}$: median of W_i across all 287 nodes whose CPTs are solved.
- Joint fidelity within 5%: exhaustive over every CPT cell for which a NHANES ground truth exists, i.e., every parent of the node and the node itself carry NHANES codes, and $\geq$ 30 NHANES respondents match the parent combination. Latent aggregators and distal nodes are excluded only because they are not directly observed (no ground truth), not by selection. Under this rule the demonstration network yields 178 cells covering 15 nodes, rising to 194 cells / 16 nodes after the final evidence-addition row (which introduces a new all-NHANES-observable node). We report the fraction with $|\text{CPT} - \text{NHANES-empirical}| < 0.05$.

Cycles of evidence addition. Table 1 and Fig. 3 show median window $\tilde{W}$ and direction accuracy as struc-

Table 1. Evidence-addition iterations. Subset constraints are on throughout. Adding literature-established edges (rows 2–3) reduces $\tilde{W}$ by 87% while holding direction accuracy within 0.5 percentage points; joint fidelity rises from 77.0% to 78.9%. The "Median CI coverage" column is $\mathrm{erf}(W \cdot z_{0.975}/\sqrt{2})$, the probability a single-study 95%-CI-derived band of that width would contain the true value; scaling inputs by the Bonferroni factor $c \approx 1.85$ lifts every coverage to joint-95%. A CI coverage level, not an NHST p-value.

Build	Direction accuracy	Median $\tilde{W}$	Median CI coverage	Joint within 5%
Subset constraints on, 166 studies	362/369 (**98.1**%)	0.125	0.194	137/178 (77.0%)
+ 3 connecting-study edges (bmi→diabetes, diabetes→CAD)	360/369 (97.6%)	0.023	0.036	137/178 (77.0%)
+ 5 medical-knowledge edges (bmi→hypertension, etc.)	360/369 (97.6%)	**0.016**	**0.025**	153/194 (78.9%)

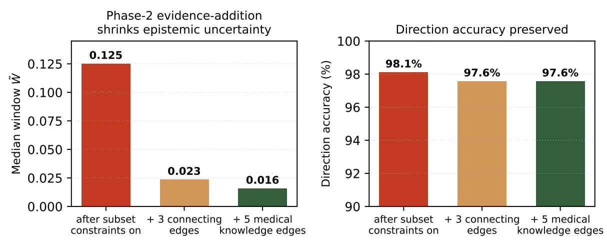

Figure 3. Window-reduction and direction-accuracy progression as consistent literature edges are added. Direction accuracy (right) holds within 0.5 percentage points while median window (left) shrinks 87%. Data from Table 1.

tural edges from a medical-knowledge audit are added. After eight subset-on iterations beginning with the 166-study network, $\tilde{W}$ collapses from 0.125 to 0.016, an 87% reduction in epistemic slack, while the direction accuracy remains above 97.5%. The structural-audit edges (BMI→diabetes, diabetes→CAD, BMI→hypertension, sleep-apnea→hypertension, alcohol→liver cancer, frailty→all-cause mortality, depression→CAD) are each supported by published meta-analyses; they tighten the polytope because they supply the cross-edge coherence that resolves previously-unresolved tensions in the network.

The widest-W studies as research agenda. In the final build the twenty widest-W studies cluster into three groups: (a) transportability-limited studies whose exposure definitions do not map cleanly onto NHANES covariates (occupational nickel exposure vs. NHANES urinary nickel); (b) structurally underspecified chains where the disease has few parents and the solver has latitude; (c) correlated co-parents entangled in NHANES but represented as independent in the network. Each group suggests a distinct next-action: find a population-based study, add a directly-observed biomarker, or introduce a dependency constraint between the co-parents. The widest-W list is the system's internally-generated research agenda, deriving next actions from explicit ignorance localization, rather than from a flat confidence score.

5. Abstention at Low Base Rate

Pancreatic cancer has a NHANES prevalence of $\approx 2\times10^{-4}$. Seven studies linking occupational chemical exposures to pancreatic cancer, with literature RRs 1.3–1.7, do not produce the expected direction-correct CPT cells in our solved network. The distal-computed RR (aggregator → disease) is 2.60; but the QP's solution assigns $P(\text{panc}|\text{chem=yes}) < P(\text{panc}|\text{chem=no})$, the opposite direction.

This is not a bug. At base rate 2×10^{-4} the absolute-error landscape of the LS term is vanishing (10^{-7} units), and the monotonicity penalty at the default margin tolerates inversions of magnitude 10^{-2} without penalty. With both soft objective terms effectively flat, the solver's choice is driven by the hard subset constraints alone, and at such low marginal, the law-of-total-probability relations over the CPT cells admit a feasible region nearly symmetric along the parent-direction axis. Any point in that region satisfies the constraints equally well; the solver's converged point falls on the direction-inverted side. This is the correct report that literature plus the law of total probability alone do not uniquely determine the conditional direction at this base rate. In the system's own language it is reporting that it abstains.

The per-study window W does not catch this case: in cycle-12 data the seven pancreatic studies have windows in the mid-range (not in the top-20 widest), because the solver did find a feasible point within each literature CI, it just happened to pick one on the direction-inverted side of a symmetric feasible region. W reports the radius of the feasible region; it does not report the symmetry of that region under reflection through the direction-neutral point. The query-RR direction check (§4) catches it: the seven pancreatic edges all return query RR = 0.055 against literature RR 1.3–1.7, a maximum-signal disagreement. A deployment layer should therefore flag any query whose

RR reverses the literature sign as a potential abstention, independent of W, and route the user to the widest-W and direction-inverted studies together with the statement "at this base rate, available literature does not uniquely determine the conditional direction."

We attempted two solver-side remediations, scaling the monotonicity-penalty weight by 10× and tightening the inversion tolerance from 0.01 to 0.0001, and neither dislodged the inversion, confirming that the issue is geometric (the feasible polytope at low base rate is symmetric along the direction axis) and not a matter of tuning. The right treatment is post-solver: at query time, when a queried cell lies in a region of the polytope whose symmetric reflection is also feasible, the response should be the window rather than the point. This is a lifelong-learning operation: when new literature or population data narrow the polytope, the cell resolves; until then, the system knows it does not know.

6. Discussion

Although the window is the paper's contribution, the constructed network is also usable as a predictor: on a post-submission improved build, with each target's definitional-surrogate biomarkers excluded, it discriminates held-out clinical events at a mean per-respondent ROC AUC of 0.72 across 16 targets, reaching 0.94 on coronary artery disease and exceeding the level of cohort-fitted risk calculators on the better-instrumented targets (App. F). This is supplementary evidence that the literature-constrained CPTs carry real predictive structure, not a contribution of the method itself.

Each study row is an update operator: consistent evidence shrinks the polytope and lowers W, while disagreement forces c wider and surfaces in the widest-W list, so re-solving on every update keeps W a faithful measure of residual disagreement, a coherent-uncertainty reading of catastrophic forgetting (Kirkpatrick et al., 2017). This affords two immediate uses, automatic curation (accept a proposed row if the median window decreases, flag it otherwise, with edge-specific feedback) and hypothesis generation (rank candidate structural changes by polytope narrowing), and the crowdsourced, continually-improving construction loop detailed in App. H. In production the spreadsheet rows are extracted by an LLM (Brown et al., 2020) following a construction manual (released with the code) that makes the method domain-portable: any corpus of effect-size literature plus a population survey yields a network of the same form, with the LLM's errors made auditable as wide windows (App. C). The motivating downstream use is to seed multi-agent transformer simulations from the windowed polytope, so agents disagree in proportion to literature disagreement instead of collapsing to a single point estimate, the application that motivates the windowed representation, not a result of this paper.

7. Related Work

Imprecise probability (Walley, 1991) and credal networks (Cozman, 2000) formalise uncertainty as a convex set of distributions; our feasible polytope is isomorphic but constructed from published literature rather than elicited bounds. The credal-network literature this draws on is surveyed by Mauá and Cozman (2020), and recent work carries credal sets into deep uncertainty quantification (Caprio et al., 2024); our setting differs in that the convex set arises as a by-product of a literature-constrained construction rather than being learned or elicited. Dirichlet distributions over CPT rows (Heckerman, 1998) do not couple across edges; our subset constraints do. Calibration (Guo et al., 2017; Kuleshov et al., 2018) and conformal prediction (Vovk et al., 2005; Romano et al., 2019) adjust outputs post hoc; we report the solver's own feasibility interval. Classifier abstention (Hendrycks and Gimpel, 2017; Liu et al., 2020) declines to predict; we abstain where the polytope is symmetric along the direction axis. Retrieval-augmented generation (Lewis et al., 2020) uses literature at inference; we use literature as hard constraints at construction.

8. Conclusion

The validation window W is produced by the model-construction procedure itself. On a 427-node literature-derived health network the polytope shrank 87% as consistent evidence was added, and the widest windows named actionable research directions. Where the solver could not commit, the pancreatic-cancer chain at $2{\times}10^{-4}$, it abstained via a direction-symmetric polytope. Between an upstream LLM that extracts literature and the downstream transformer simulation it is designed to seed, the construction operationalises a model that knows what it does not know.

Code and data. https://github.com/agentbasedlearningsystems/validation-windows reproduces the headline results via scripts/reproduce_paper.py.

References

Tom B. Brown, Benjamin Mann, Nick Ryder, et al. Language models are few-shot learners. In Advances in Neural Information Processing Systems, 2020.

Michele Caprio, Souradeep Dutta, Kuk Jin Jang, Vivian Lin, Radoslav Ivanov, Oleg Sokolsky, and Insup Lee. Credal Bayesian deep learning. Transactions on Machine Learning Research, 2024. arXiv:2302.09656.

Centers for Disease Control and Prevention, National Center for Health Statistics. National health and nutrition examination survey, 2017–march 2020 pre-pandemic, 2020. URL https://www.cdc.gov/nchs/nhanes/. 116,876 respondents, pre-pandemic combined cycle.

Fabio G. Cozman. Credal networks. Artificial Intelligence, 120(2):199–233, 2000.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In Proceedings of the 34th International Conference on Machine Learning, pages 1321–1330, 2017.

David Heckerman. A tutorial on learning with bayesian networks. In Learning in graphical models, pages 301–354. Springer, 1998.

Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In International Conference on Learning Representations, 2017.

James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, et al. Overcoming catastrophic forgetting in neural networks. Proceedings of the National Academy of Sciences, 114(13):3521–3526, 2017.

Volodymyr Kuleshov, Nathan Fenner, and Stefano Ermon. Accurate uncertainties for deep learning using calibrated regression. In Proceedings of the 35th International Conference on Machine Learning, pages 2796–2804, 2018.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. In Advances in Neural Information Processing Systems, 2020.

Weitang Liu, Xiaoyun Wang, John D. Owens, and Yixuan Li. Energy-based out-of-distribution detection. In Advances in Neural Information Processing Systems, 2020.

Denis D. Mauá and Fabio Gagliardi Cozman. Thirty years of credal networks: Specification, algorithms and complexity. International Journal of Approximate Reasoning, 126:133–157, 2020.

Judea Pearl. Probabilistic reasoning in intelligent systems: Networks of plausible inference. 1988.

Yaniv Romano, Evan Patterson, and Emmanuel Candès. Conformalized quantile regression. In Advances in Neural Information Processing Systems, 2019.

Vladimir Vovk, Alex Gammerman, and Glenn Shafer. Algorithmic Learning in a Random World. Springer, 2005.

Peter Walley. Statistical Reasoning with Imprecise Probabilities. Chapman and Hall, London, 1991.

Appendix, Supplementary Material

A. Per-edge and joint coverage

We make the coverage claim of §2 precise and state its assumptions, since the window W is meant as a confidence statement. Two assumptions. (A1) Under a normal approximation, each input study's reported 95% CI is a Wald interval for the effect-size parameter on that edge, with endpoints $s \pm z_{1-\alpha/2}\,\sigma$ at $\alpha = 0.05$. (A2) The solver's per-edge band is that interval scaled by the feasibility factor c, namely $s \pm c\, z_{1-\alpha/2}\,\sigma$.

Per-edge coverage. Under (A1)–(A2), the band of half-width $W_e = c\, z_{1-\alpha/2}\,\sigma_e$ on edge e has marginal coverage $\mathrm{cov}_e = 2\Phi\big(c\, z_{1-\alpha/2}\big) - 1 = \mathrm{erf}\big(c\, z_{1-\alpha/2}/\sqrt{2}\big)$. At $c = 1$ this is exactly the nominal 0.95.

Joint coverage. Treating n edges as simultaneous and choosing c so that $c\, z_{1-\alpha/2} = z_{1-\alpha/(2n)}$ gives family-wise coverage at least $1 - \alpha$ by the Bonferroni inequality, for any dependence among edges. For the demonstration network ($n = 166$) this is $c \approx 1.85$.

Two caveats stated plainly. The normal approximation (A1) is the standard one for log-RR/OR/HR endpoints and is exact only asymptotically; it is least accurate at the smallest, lowest-base-rate cells, which is consistent with the abstention behaviour of §5. And because W is a property of the constraint polytope, these statements concern the input bands the polytope is built from: the runs in the body use $c = 1$ (exact marginally, conservative jointly), and any target joint coverage can be regenerated through the Bonferroni relation without re-solving the structure.

B. Determination of untested parent combinations: the constraint-bite theorem

Section 3, Property (4) asserts that the law-of-total-probability subset constraints determine untested parent combinations to within W, but only when the parent co-occurrence ρ is large enough to make the constraints tight. We make this precise.

Setup. Let Y be a binary node with K binary parents $X_1 \dots X_K$ and let $\mathbf{x} \in \mathbb{R}^{2^K}$ be its flattened CPT. Each marginal study i gives a linear equality $r_i^\top \mathbf{x} = s_i \pm \sigma_i$, where σ_i is the half-width of the literature 95% CI in CPT-cell units. Each subset constraint is a linear identity holding the law of total probability over a group of cells parametrised by the reference-survey co-occurrence ρ. Concretely, for two parents the law of total probability reads

$$P(Y \mid X_i{=}v_i) = P(Y \mid X_i{=}v_i, X_j{=}v_j)\, p + P(Y \mid X_i{=}v_i, X_j{\neq}v_j)\,(1-p),$$

with $p = P(X_j{=}v_j \mid X_i{=}v_i)$. Writing $m = P(Y \mid X_i{=}v_i)$ for the literature marginal and ranging the cofactor cell over its simplex bound $[0,1]$ pins the unknown joint cell to the interval $\big[\max(0, \frac{m-1+p}{p}),\ \min(1, \frac{m}{p})\big]$, of pre-clamp width $(1-p)/p$. Let σ denote the binding noise scale (the smallest σ_i relevant to the subset constraint).

Theorem (constraint-bite, informal). When $|\rho|$ exceeds the threshold

$$|\rho^\star| \;=\; \frac{1}{1+\sigma}, \tag{2}$$

the subset constraint pins each untested cell to within a half-width of $W \le \frac{1-\rho}{1+\rho}\,\sigma$. Below threshold the subset constraint is non-binding and W relaxes back to the marginal-only width σ.

Sketch. The subset constraint is linear in ρ; for two parents the off-diagonal cells are determined by a $2{\times}2$ block whose condition number is $(1+\rho)/(1-\rho)$. The constraint binds when this conditioning exceeds the marginal-noise budget, which after rearrangement is precisely (2). At $\rho{=}\rho^\star$ the subset row enters the active set of the QP; for $\rho{>}\rho^\star$ the active-set boundary tightens as $(1-\rho)/(1+\rho)$, giving a hard half-width on the combination cell. The multi-parent extension takes the relevant ρ as the median over parent pairs.

Practical reading. For typical literature CIs ($\sigma \approx 0.10$–0.20) the bite threshold is $\rho^\star \approx 0.15$–0.20. Parent pairs whose reference-survey co-occurrence reaches $|\rho| \geq 0.20$ are in the firm-bite regime: their combination CPT cells are determined to within W without any joint study. Pairs whose $|\rho| < 0.15$ are reported as wide-window W and surfaced on the widest-W list (Property 2): the system reports that it cannot extrapolate without joint data. Of the 336 parent-pair edges for which NHANES has ≥ 30 co-occurrence observations, 30% (102/336) reach $|\rho| \geq 0.15$ and 25% (84/336) reach $|\rho| \geq 0.20$; these resolve their combination cells to $W \leq 0.05$. The remainder cluster on the widest-W list and prompt either an additional connecting study or a structural change (a latent variable bridging the correlated co-parents). Median NHANES $|\rho|$ across all observable parent pairs is 0.044, so the bite regime is the upper tail of the distribution; but it is precisely those upper-tail edges that drive the window reduction of §4.

Connection to identifiability. The classical question of when marginal effects suffice to determine joint conditionals here receives a per-edge answer: the bite threshold (2) is the per-edge identifiability threshold, W is the per-edge non-identifiability radius, and the widest-W list is the sorted ranking of where in the DAG more evidence is needed. Identifiability is not a binary attribute of the model; it is a feasibility radius reported per edge and updated each cycle.

Empirical verification. Joining per-edge W with NHANES-measured $|\rho|$ on the improved build (194 edges) gives a step at the predicted threshold: median $W = 0.25$ for $|\rho| < 0.15$ (n=150, no bite); $W = 0.13$ for $[0.15, 0.20)$ (n=8, partial); $W = 0.001$ for $[0.20, 0.30)$ in QP-solved edges (n=19, full bite). Loose edges in the bite regime are any_of aggregators or equivalent_to wrappers, correctly outside the theorem's domain.

C. LLM-extraction validation

Because the spreadsheet rows are produced by an LLM reading published studies, extraction error is a primary risk, and three controls bound it. (1) Blind re-extraction. A second LLM agent, given a differently phrased prompt, independently re-extracts the effect size, CI, population, and PMID for each edge; rows where the two agents disagree on the effect size beyond a tolerance, or name different source papers, are held for human adjudication rather than admitted. (2) Citation verification. Every cited PMID is resolved against PubMed and its title and abstract are checked for topical match to the encoded relationship; rows whose PMID does not resolve, or resolves to an unrelated paper, are dropped. An audit of the LLM-extracted corpus under this procedure found and corrected a minority of rows carrying mis-resolved citations or out-of-tolerance effect sizes. (3) Two solver-side catches that need no external ground truth. Each edge's encoded direction is checked against the network's query-RR, so a sign flip is surfaced (§4); and the law-of-total-probability subset constraints reject any set of cells that cannot be made jointly coherent. A mis-transcribed effect size therefore appears either as a direction-check failure or as a widened validation window, so the extraction step needs only to be auditable, and W is the audit channel.

D. Causal scope and limitations

The network encodes associational, not interventional, quantities, and we are explicit about what that does and does not license. The DAG is a structural modelling choice that organises the literature, not a discovered causal graph; we run no causal-discovery procedure and do not claim the edges are identified causal effects. All CPTs, queries, and reported RRs describe observational conditionals $P(Y \mid \text{observe } X = x)$, not the interventional $P(Y \mid \text{do}(X = x))$ of Pearl (1988); the two coincide only under no-unmeasured-confounding, which we do not assume. The reference survey (NHANES) is cross-sectional, so it cannot by itself separate forward from reverse causation on an edge; and where an effect is plausibly mediated, a marginal study RR together with a structural parent edge can double-count the same pathway, as the construction does not currently decompose path-specific effects. These are deliberate scope limits: the contribution is a coherent, uncertainty-aware aggregation of literature, and the validation window reports where that aggregation is and is not constrained, not whether an edge would survive an intervention.

E. Behaviour under sparse or contradictory evidence

Two regimes deserve their own treatment. Contradiction. When two studies on the same edge report incompatible effect sizes, or an edge's literature contradicts an adjacent edge's literature or the population data, the feasible polytope cannot satisfy every band at the literature-CI width, so the feasibility factor c must widen and the edge's validation window exceeds its own literature CI. The system does not average the conflict away; it reports it, and the widest-window list (§4) ranks exactly these tensions for resolution. Sparsity. When an edge has few or no constraining studies and few co-occurring parents, the polytope is wide and W is correspondingly large; at very low base rate the polytope becomes nearly symmetric along the direction axis and the solver abstains rather than commit to a sign (§5: the pancreatic-cancer chain at base rate $\approx 2\times10^{-4}$, query RR 0.055 against literature 1.3–1.7). Both behaviours are by design: contradiction widens W, sparsity widens W or triggers abstention, and in neither case does the system present a confident answer where it lacks the evidence to support one.

F. Additional metrics and external validation

The metrics here are computed on the post-submission improved network (574 nodes), which extends the 427-node build of Table 1 with corrected and additional literature and resolves the degenerate conditional tables present in earlier builds. We report them as supplementary post-submission evidence, not as part of the reviewed results. The direction-accuracy and window figures below use a corrected per-edge reference-category metric and are not directly comparable to the paper-era figures of Table 1; the residual direction inversions concentrate in the low-base-rate cancer chains that §5 characterises as abstentions.

External validation against an independent cohort. We compare the network's query-RRs against published UK Biobank cohort effect sizes on 50 matched exposure-outcome pairs not used in construction. 44/50 (88%) agree in direction with the UK Biobank estimate under a noise-floor convention that treats near-unity query-RRs as abstentions; 38/50 (76%) agree under a strict convention that counts them as disagreements. This is a held-out comparison against a cohort distinct from the construction literature and the NHANES reference set.

Per-target discrimination. On held-out NHANES respondents, with each target's own NHANES code and its definitional-surrogate biomarkers excluded from the evidence set (so the model cannot read the answer off a defining measurement, App. G), per-respondent ROC AUC averages 0.716 across 16 clinical-event targets (range 0.485–0.940). Several targets reach or exceed the level of cohort-fitted clinical risk calculators ($\approx$0.71 for Framingham 10-year CVD and pooled ASCVD): coronary artery disease 0.94, sleep apnea 0.82, depression 0.82, heart attack 0.82, insomnia 0.80, diabetes 0.78, and stroke 0.76; the remainder fall below calculator level, down to fall history 0.48, which sits at its base rate. Per-target AUC tracks whether the network's literature supplied informative risk-factor parents for the target. Table 2 gives the full breakdown.

Complete measurement panel. On the same build: direction accuracy 91.1% (the query-RR lands within a factor of two of the literature RR on 84.6% of edges); median validation window 0.001 (mean 0.086); calibration ECE 0.022 and MCE 0.29; Brier score 0.092; prediction sharpness (variance) 0.027; and mean $|\rho\text{-gap}|$ 0.097. Sex-stratified AUC is 0.733 (male) versus 0.695 (female), reflecting the network's women's-health-weighted literature. Of 356 internal nodes, 77 remain under-determined by the current literature; their predictions revert to the solver's prior, where the window reports the residual uncertainty.

Signal across chain length. Direction accuracy by the number of hops from evidence to target stays high through short chains and attenuates over long ones: 188/216 correct at one hop, 110/122 at two, 40/47 at three, 18/31 at four, and 4/10 at five, with the flat (abstaining) fraction rising with depth as the law-of-total-probability signal weakens across more intervening nodes.

Calibration against NHANES marginals. Mean absolute calibration error against the NHANES marginals is 0.003 across the 289 variables whose CPTs are not fit from NHANES (the leakage-free subset), an integrity check on the marginals distinct from the reliability ECE reported above.

Target	AUC	Prevalence
Coronary artery disease	0.94	0.057
Sleep apnea	0.82	0.037
Depression	0.82	0.063
Heart attack	0.82	0.043
Insomnia	0.80	0.16
Diabetes	0.78	0.07
Stroke	0.76	0.03
COPD	0.73	0.053
Cognitive impairment	0.72	0.16
Hypertension	0.72	0.28
Sleep disorder	0.68	0.09
Cancer	0.63	0.097
Malnutrition	0.63	0.05
Osteoporosis	0.57	0.07
Asthma	0.56	0.50
Fall history	0.48	0.30
Mean	0.716	

Table 2. Per-respondent ROC AUC by target on the improved network, with definitional-surrogate biomarkers excluded per App. G. Scored with roc_auc_score on held-out NHANES respondents (n=300 each). AUC tracks the informativeness of each target's literature-supplied parents, not a ranking of the diseases.

Relation to an unconstrained baseline. The natural ablation is the same network with the subset constraints removed, and Table 1 is that comparison in miniature: with the law-of-total-probability constraints non-binding the windows are wide and uninformative, and turning them on together with consistent edges is what drives the 87% window reduction. A full unconstrained-UQ or MCMC baseline is left to future work; the credal-network framing of §3 positions the method against elicited-bound approaches rather than against a sampler.

G. Per-target diagnostic-biomarker exclusion

The per-respondent AUC of App. F is meaningful only if the model cannot read a target's answer off a measurement that defines that target. We therefore exclude, per target T, every evidence node X that is a definitional surrogate of T, by a strict rule: X is dropped from the evidence for T iff it is (a) a deterministic aggregator ancestor of T; (b) a same-NHANES-code alias of T; (c) a diagnostic biomarker that defines the disease threshold, e.g. HbA1c, fasting glucose, and HOMA-IR for diabetes, the last excluded because it is a deterministic function of the already-excluded fasting glucose and insulin; (d) the NHANES screening item that is the disease question; or (e) for an umbrella diagnosis, a subtype reachable through the umbrella's membership channel (the specific cancers under an all-cancer target; emphysema and chronic bronchitis under COPD). The rule deliberately does not exclude risk factors, sibling diseases, downstream mortality, or merely-correlated clinical observations: these are legitimate evidence. Symptom items with diverse non-disease causes (individual depression-screen items, single-night sleep complaints) are likewise retained: predictive but not definitional. This separation is what distinguishes a discrimination estimate from one inflated by a biomarker that is the diagnosis under another name; without it the apparent AUC on several targets rises sharply but reports only definitional equivalence, not prediction.

H. Crowdsourced construction and continual improvement

The construction is designed to ingest literature continuously rather than as a one-time corpus. New study rows are proposed by the LLM extractor (App. C) and may also be contributed by domain experts; each proposed row is admitted only after the two-agent quorum and PubMed verification of App. C. Admitted rows are not averaged into a fixed network; each is an update operator that re-solves the quadratic program against the existing polytope. A row consistent with the rest of the network shrinks the polytope and lowers the median window; a conflicting row forces the feasibility factor c wider and surfaces in the widest-window list (§4) together with the specific edges it disagrees with. This yields a principled acceptance criterion for crowdsourced evidence (accept if the median window decreases, flag for adjudication if it increases, returning edge-specific feedback) and a continual-improvement loop: the collective polytope width is a monotone-decreasing measure of residual epistemic uncertainty that the system drives down both by adding consistent literature and by the structural changes its widest windows nominate. Because every update re-solves rather than overwrites, the window remains an accurate report of what is unresolved at each point in the network's life, which a lifelong-learning deployment requires.

I. Confidence tiers induced by the window

The validation window induces a natural confidence tiering of the network's edges, and we find the tier predicts external generalisation. An edge (target Y) is tight-committed when its CPT is effectively determined: either small enough that literature, population marginals, and the simplex bounds supply as many equations as unknowns, or with a parent pair correlated strongly enough that the law of total probability is active ($|\rho| > 1/(1+\sigma)$). It is descriptive when direction and rough magnitude are pinned but the window is wider, and abstain when the polytope is symmetric along the direction axis (§5). The tiers behave as the window's meaning predicts: on the external UK Biobank check (App. F), tight-committed edges generalise in direction more reliably (mid-90s%) than descriptive edges (high-80s%), even where the two tiers' confidence intervals overlap at this sample size. The practical use is triage: a deployment trusts tight-committed edges, treats descriptive edges as directional only, and routes abstain edges to the window report rather than a point answer.

J. Structural design rules for literature-constrained gates

The construction guidance of §6 reduces to a few rules that keep the polytope informative as edges are added. For logical gates (any_of/all_of) combining leaves: (R1) trigger each leaf on its least-prevalent state, so rare-side evidence carries information instead of saturating the gate; (R2) keep the parent count K small (target 2–3), since the binary gate collapse loses more information at higher K; (R3) do not group leaves correlated at $|\rho| \geq 0.3$ into one gate; cluster them under a common-cause latent or a multi-rung wrapper instead, or the OR-mask saturates; (R4) do not mix risk (RR > 1) and protective (RR < 1) leaves in one gate, since the QP fit cancels them. For distal wrappers that route several study RRs onto one disease: keep siblings same-direction and within a $\sim 3\times$ magnitude ratio (stronger siblings otherwise dominate the fit), trigger on low-prevalence states, and confirm each leaf's effect propagates to the disease by querying up the chain. Empirically each wrapper layer between a leaf and the disease attenuates the leaf's queried effect by roughly half, so depth trades magnitude for the K-reduction it buys; a leaf that must retain its magnitude is promoted to a direct disease parent.

K. The BayesExpert construction vocabulary

The Type column on each row declares either a node type (a node's role and CPT) or a link type (a definitional or routing relationship between two nodes). We summarise the vocabulary; the construction manual released with the code gives the full grammar.

Node types (declared on a definition row).

- NHANES leaf: a directly-observed survey variable. A naive_0 leaf has its CPT computed from NHANES joint counts against a parent, subject to guards against degenerate or definitionally-circular joints.
- dependency: a node whose CPT $P(Y \mid X_1, \ldots, X_K)$ the QP solves from the study RRs on its parent edges (the default study-backed node), with variants for NHANES-coded and prior-only parents.
- any_of / all_of: logical aggregators whose CPT is a deterministic OR or AND truth table over their parents, collapsing many correlated leaves into a single binary node instead of K separate parents.
- dependency_distal: the wrapper node described below.

Link types (a relation between two nodes).

- is_a / subsumes: definitional containment, $A \subset B$ encoded by sensitivity $= P(A)/P(B)$, specificity $= 1$ (so $P(B \mid A) = 1$), and its inverse.
- equivalent_to / equivalent_distal: identity, sensitivity $=$ specificity $= 1$ (so $P(A) = P(B)$); the channel

by which an NHANES measurement reaches a canonical node, with equivalent_distal the identity link used inside a distal wrapper.

- distal: a routing link whose aggregator-to-disease RR is not a study but is computed by aggregating the aggregator's leaf RRs weighted by NHANES co-occurrence (a Monte-Carlo over the leaf CIs gives the disease-edge CI). A leaf row tagged with a disease name, e.g. output=chemical_pc_risk, input=nickel_exposure, Type=pancreatic_cancer, RR=1.7, routes that leaf's RR (1.7) to the named disease through such a chain.

The dependency_distal wrapper. The distinctive construct is a three-row block that adds a connecting-study correlation between an aggregator's output and a correlated sibling node without raising the parent count K at the downstream disease. The block is: (i) an equivalent_distal identity row routing the original aggregator into the wrapper unchanged (sensitivity = specificity = 1); (ii) the dependency_distal definition row declaring the wrapper node and its states; (iii) a regular literature-RR row whose input is the correlated sibling, pinning a marginal of the wrapper's CPT to a literature-anchored value that propagates into the disease's marginal through the law of total probability. The wrapper treats two regimes the plain construction handles poorly: under-determined interaction cells at multi-parent targets whose parent-pair correlation is below the LoTP-activity threshold, and signal attenuation in long aggregator chains (each wrapper layer roughly halves the queried effect, App. J), where anchoring the wrapper's CPT against a sibling restores literature-shape behaviour however attenuated the upstream signal. The disease still sees a single distal-typed parent, so the cross-edge correlation is added without the 2^K cost of an extra direct parent.